

Creativity Toolkit - First Draft

Based on [Pre-assessment of Faculty](#) 10.7.25

Collaborative

What It Looks Like

Practices to Try

1. Structured Peer Interaction
2. Assigned Roles and Shared Responsibility
3. Peer Feedback and Reflection
4. Shared Inquiry and Problem Solving
5. Helping Each Other Learn
6. Communication Skills and Kindness

Teacher Moves

Imaginative

What It Looks Like

Practices to Try

1. Playing with Possibilities
2. Visual Thinking and Mapping Ideas
3. Open-Ended Materials and Provocations
4. Brainstorming, Tangents, and "Rabbit Holes"
5. Storytelling and Creative Connection
6. Making Connections and Using Intuition

Teacher Moves

Persistent

What It Looks Like

Practices to Try

1. Growth Mindset and "The Power of Yet"
2. Learning Through Mistakes
3. Revision, Reflection, and Iteration
4. Embracing Uncertainty and Challenge
5. Encouraging Self-Regulation and Flexible Thinking
6. Modeling and Meta-Learning

Teacher Moves

Disciplined

What It Looks Like

Practices to Try

1. Drafting, Revising, and Refining

- [2. Reflection and Self-Assessment](#)
- [3. Precision, Craft, and Technical Mastery](#)
- [4. Scaffolding and Step-by-Step Learning](#)
- [5. Reflection Through Feedback and Collaboration](#)
- [6. Iteration and Reflection Beyond the Classroom](#)

[Teacher Moves](#)

[Inquisitive](#)

[What It Looks Like](#)

[Practices to Try](#)

- [1. Wonder as a Starting Point](#)
- [2. Question-Based Routines](#)
- [3. Investigating and Proving Thinking](#)
- [4. Challenging Assumptions and Exploring Perspectives](#)
- [5. Playful and Digital Inquiry](#)
- [6. Modeling and Sustaining Curiosity](#)

Collaborative

Creative learners know that great ideas grow stronger together.

When students collaborate, they listen closely, build on others' ideas, and make space for every voice. Collaboration in creative learning isn't about dividing tasks — it's about co-creating understanding. By working together through structured dialogue, shared responsibility, and kind feedback, students learn that thinking collectively leads to deeper insight and more imaginative outcomes.

"Collaboration is where creativity becomes community."

What It Looks Like

- Students exchange and build on one another's ideas during structured conversations.
- Groups take on distinct roles that promote equity, voice, and shared responsibility.
- Feedback routines emphasize listening, reflection, and kindness.

- Peers serve as coaches, questioners, and creative partners.
-

Practices to Try

1. Structured Peer Interaction

Predictable formats give all students a way to engage and be heard.

- **Think-Pair-Share / Turn & Talk / Pair and Share** – Students reflect, share with a partner, and then share their partner’s idea with the class.
 - **Dialogue Circles** – Shared conversations around topics or issues that draw on community knowledge.
 - **Round Robin Repeating / Talking Object / Written Conversations** – Students build on each other’s language to deepen understanding.
 - **Late Night Book Talk** – Partner interviews about books or projects, encouraging curiosity and connection.
-

2. Assigned Roles and Shared Responsibility

Clarity of roles builds equity and accountability.

- **Team Roles** – Each student has a defined task (e.g., timer, instruction checker, airtime controller).
- **One Marker, Two Minds / Two Brains, One Marker** – One student writes, the other thinks — they must trade often!
- **Collaborative Drawing** – One student describes an image while another draws it sight unseen.
- **Job Roles in Projects** – Students practice independence by managing project elements themselves.

- **Wipebooks for Small-Group Thinking** – One shared surface captures the team’s evolving ideas.
-

3. Peer Feedback and Reflection

Students learn to improve work — and relationships — through thoughtful response.

- **Warm, Cool, Question** – “Glow and grow” model for peer feedback.
 - **Peer Editing with Rubrics** – Students offer specific, evidence-based critique.
 - **Art Critiques with Guided Prompts** – Class reflections framed around observation and inquiry.
 - **Gallery Walks** – Peers move around to view, reflect on, and comment on one another’s work.
 - **Google Presentation Feedback Forms** – Short reflections that balance strengths and growth areas.
-

4. Shared Inquiry and Problem Solving

Students generate solutions together, learning to value diverse thinking.

- **Brainstorming Around Materials or Solutions** – Ask, “What would make this joint stand up?” or “What could we try next?”
- **Collaborative Math Talk** – Students articulate reasoning and compare problem-solving approaches.
- **Team Challenges** – Simulated stressors with reflection on group dynamics and creative problem-solving.
- **Improv Games (“Yes, and...,” One-Word Story)** – Build creative agility through collective storytelling.
- **Gallery Walk of Mathematical Work** – Students study and question each other’s problem-solving strategies.

5. Helping Each Other Learn

Peers act as guides, teachers, and cheerleaders for one another.

- **Get It, Got It, Give It** – Students who understand a topic circulate to help peers.
- **Ask a Friend, Help a Friend** – Encourages seeking peer support first.
- **Tech Tip Helpers** – Student experts assist with digital tools and troubleshooting.
- **“When Stuck, Ask a Question” Routine** – Partners prompt each other toward new perspectives.
- **Walk for Inspiration** – When blocked, students take a gallery walk to spark ideas from others’ work.

6. Communication Skills and Kindness

Creativity flourishes when collaboration is rooted in empathy and respect.

- **“Before You Speak, Think” Protocol** – Promotes kindness and mindfulness in communication.
- **Sentence Discussion Stems** – “I think...,” “I see...,” “What do you think?” to model constructive dialogue.
- **Repeating and Adding On** – Reinforces listening and building on others’ ideas.
- **Share and Shift Perspectives** – Encourage curiosity about others’ viewpoints.
- **Patience as Practice** – Name patience and encouragement as key creative behaviors.

Teacher Moves

- Establish norms of respect, listening, and inclusion early and reinforce them often.

- Use visible tools (sentence stems, feedback prompts, role charts) to scaffold collaboration.
- Model “thinking aloud” and productive disagreement.
- Reflect with students on *how* they collaborated, not just *what* they produced.
- Celebrate examples of creative interdependence — “We got there together.”

Imaginative

Creative learners play with possibilities and explore “what if?”

Imaginative thinking asks students to wonder, invent, and connect. When they make space for the unexpected — whether through sketching wild ideas, finishing half-drawn stories, or asking playful questions — students learn that creativity grows through curiosity and exploration. Teachers nurture imagination by giving permission to take risks, make connections, and imagine new ways of seeing the familiar.

“Imagination is intelligence at play.”

What It Looks Like

- Students generate multiple ideas and explore “what if” scenarios.
 - Learning experiences invite curiosity, play, and speculation.
 - Teachers use open-ended prompts, visuals, and materials that provoke wonder.
 - Mistakes and tangents are treated as fuel for creativity.
-

Practices to Try

1. Playing with Possibilities

Students engage their imagination when they’re encouraged to explore the unexpected.

- **Think Outside the Box Thursdays** – Students complete an unfinished image (like half an ice cream cone) in a creative way.
- **Finish the Drawing / Finish the Story** – Students transform a partial drawing or story opening into something new.
- **“What if...” Questions** – Speculative prompts: *What if the sun exploded? What if dogs could read minds?*
- **Yes, and... Rule** – Build ideas collaboratively without judgment during the creative phase.
- **Worst Idea Generation** – Students brainstorm “as many terrible ideas as possible” before refining.
- **Evolving Constraints** – Provide minimal or shifting parameters and let students build around them.
- **Improv-style Thinking** – Encourage quick, humorous, or exaggerated scenarios to loosen up creative risk-taking.

● *Purpose:* Normalize exploration and humor as tools for innovation.

2. Visual Thinking and Mapping Ideas

Drawing, mapping, and visual note-taking help make invisible ideas visible.

- **Mind Maps / Spider Webs / Concentric Circles** – Central ideas branch into connected thoughts or related terms.
- **Rotating Group Mind Maps** – Each group adds to another’s map to expand thinking.
- **Sketchnotes** – Combining words and visuals to capture and connect ideas.
- **I’m a Tree Activity** – Students visualize roots (foundations), trunk (self), and leaves (growth needs).
- **Wonder Wall** – Ongoing list of students’ questions about a topic.

- **Character Motivation Mapping** – Students explore “why” characters act the way they do.
- **Mind Maps for Vocabulary** – Connecting new words to related ideas or root meanings.

● *Purpose:* Support imagination through visualization, organization, and connection-making.

3. Open-Ended Materials and Provocations

Physical and conceptual materials without a single “right” answer invite exploration.

- **Loose Parts Play** – Natural or found objects (stones, wooden pieces, gems) for dramatic play or building.
- **Provocation Displays** – Tables or spaces with open-ended materials and a central “wondering” question.
- **Design Thinking Projects** – Students invent systems or solve problems (e.g., designing life on another planet).
- **Mathematical Manipulatives** – Using objects to explore patterns, shapes, and relationships.
- **Evolving Constraints** – Start with minimal rules, then adjust as ideas develop.
- **Creating Maps of New Places** – Students invent new landscapes to explore real-world geography and resources.

● *Purpose:* Let imagination emerge from tactile exploration and authentic curiosity.

4. Brainstorming, Tangents, and “Rabbit Holes”

Imaginative classrooms celebrate divergent thinking and curiosity-driven inquiry.

- **Group Brainstorms** – Students record every idea first, sort later.

- **Stream-of-Consciousness Brain Dumps** – Write freely without editing, then highlight the best ideas.
- **Going Down Rabbit Holes** – Follow student tangents to see where curiosity leads.
- **Service Learning Brainstorms** – Generating ideas for social justice or community projects.
- **Using Padlet, Canva, or Visual Tools** – Capture and share ideas in multimedia forms.

● *Purpose:* Encourage fluency of ideas — quantity before quality.

5. Storytelling and Creative Connection

Narrative invites students to imagine possibilities and make meaningful links across subjects.

- **Storytelling in Math** – Creating characters or scenarios to explore abstract concepts (e.g., *the dragon of prime numbers*).
- **Reading “Big, Wild, Improbable Ideas”** – Start units with imaginative literature to spark curiosity.
- **Character Motivation Exploration** – Deepen writing through empathy and emotional reasoning.
- **Connecting Books and Stories** – Linking characters, plots, or themes across different texts.
- **Imaginative Writing Prompts** – “Have you ever seen...?”, “Does this feel familiar?”, or “How would this sound or look?”

● *Purpose:* Use storytelling to humanize abstract ideas and exercise creative empathy.

6. Making Connections and Using Intuition

Imaginative thinkers spot patterns and see relationships others might miss.

- **Brainstorm Multiple Solutions** – “Come up with one idea to try and one as a backup.”
- **Connecting Tangents to Core Ideas** – Encourage students to link off-topic thoughts back to central themes.
- **Mathematical Exploration Challenges** – Use digits and operations in as many ways as possible to reach a target number.
- **Vocabulary Connection Activities** – Relating new words to sound, feeling, or look-alikes (“*What does this remind you of?*”).
- **Respect and Encourage Intuitive Leaps** – Validate guesses, gut feelings, and unconventional connections.

● *Purpose:* Value intuition and connection-making as signs of creative insight.

Teacher Moves

- Model curiosity — ask playful, “What if...?” questions yourself.
- Allow time for idea fluency before refinement or judgment.
- Use visuals, metaphors, and sensory cues to spark imagination.
- Honor tangents — they’re often gateways to deeper understanding.
- Frame constraints as opportunities for invention, not limitations.
- Encourage joy, wonder, and a touch of silliness in creative work.

Persistent

Creative learners stick with challenges, take risks, and learn from mistakes.

Persistence means daring to stay in the struggle — to revise, adapt, and keep going when things get hard or uncertain. Teachers cultivate persistence by normalizing failure, modeling perseverance, and designing experiences where struggle is a natural and necessary part of learning. Students discover that persistence isn’t about perfection; it’s about patience, resilience, and belief in their own capacity to grow.

"We can do hard things."

What It Looks Like

- Students view mistakes as learning opportunities, not failures.
 - Teachers model their own persistence and thinking aloud.
 - Activities include productive struggle and revision cycles.
 - Classrooms emphasize growth, reflection, and courage in the face of uncertainty.
-

Practices to Try

1. Growth Mindset and "The Power of Yet"

Help students see ability as flexible and intelligence as expandable.

- **Growth Mindset Language** – "Your brain gets stronger when you work it."
- **The Magical Yet** – Shift "I can't do this" to "I can't do this yet."
- **Fantastic Elastic Brain (Read Aloud)** – Visualize how persistence strengthens neural pathways.
- **Class Mottos** – "We can do hard things." "Mistakes help us grow."
- **Visual Reminders** – Posters or anchor charts reinforcing persistence language.
- **Praise for Effort, Not Perfection** – "Persistent praise for failure" reframes struggle as progress.

● *Purpose:* Build emotional resilience and normalize challenge as part of the creative process.

2. Learning Through Mistakes

Students gain confidence when teachers model and celebrate imperfection.

- **“My Favorite Oops” Routine** – Highlight a common mistake as a class learning opportunity.
- **Modeling Mistakes** – Teachers make their own errors visible and narrate how they problem-solve.
- **Show Work-in-Progress** – Share examples of student projects that show effort, revision, and growth rather than polish.
- **Celebrate Failure** – Offer low-stakes chances for experimentation and learning from errors.
- **Art & Pottery Challenges** – How can a “broken pot” or “flawed glaze” become part of the design?
- **Bridge Building Projects** – Designed to fail first; revision and redesign are built into success.

 *Purpose:* De-stigmatize mistakes and build comfort with iterative thinking.

3. Revision, Reflection, and Iteration

Persistence grows when students see improvement over time and value the process, not just the product.

- **Revision Cycles** – “What else can you try?” Encourage multiple drafts, prototypes, or reattempts.
- **Iteration Checkpoints** – Early feedback points before the final submission.
- **Revising for a Specific Goal** – Focus on one improvement area rather than perfection.
- **Documenting Growth** – Students compare early and later work samples; reflect on progress.
- **Pumpkin Stopper Redesigns** – After failure, students analyze why and redesign based on new evidence.
- **Persistent Praise for Trying Again** – Celebrate resilience and curiosity, not just outcomes.

● *Purpose:* Make persistence visible and measurable through cycles of effort and reflection.

4. Embracing Uncertainty and Challenge

Students strengthen persistence when they face situations that require patience and flexible thinking.

- **Open-Ended Problems** – Tasks with multiple solutions or no single “right” answer.
- **Debates on the Opposite Side** – Arguing against one’s own view builds adaptability.
- **Waiting Before Asking for Help** – “Try for 3 minutes before raising your hand.”
Encourages independence.
- **Coding Projects** – Natural opportunities for trial, error, and debugging.
- **End on a Cliffhanger** – Practice tolerating uncertainty in reading or inquiry-based lessons.

- **Complex BIG Lab Projects** – Designed so success requires revision and collaboration.
- **Pomodoro Method** – Work intensely, then pause for reflection and reward.

● *Purpose:* Teach endurance and calm in moments of uncertainty or frustration.

5. Encouraging Self-Regulation and Flexible Thinking

Persistence isn’t only about grit — it’s also about emotional balance and perspective.

- **Calming and Encouragement Techniques** – Coach students through stress or frustration.
- **Teaching “Size of the Problem”** – Distinguish small vs. big challenges and appropriate responses.

- **Flexible Thinking Exercises** – Generate multiple strategies for solving the same issue.
- **Solutions Tool Belt** – Students share personal strategies and borrow from peers.
- **Executive Functioning Heroes** – Help students identify inner “tools” for focus and perseverance.
- **Iteration in Pottery or Art** – Checklists and mindfulness techniques help maintain focus and grounding.

● *Purpose:* Equip students with self-awareness and adaptability for sustained effort.

6. Modeling and Meta-Learning

Teachers’ persistence becomes a model for students’ own.

- **Think-Aloud Modeling** – Narrate your own struggle (“I’m not sure yet—let’s see what happens if...”).
- **Use Natural Moments to Demonstrate Perseverance** – Turn real-time challenges into teachable examples.
- **Encourage Disagreement and Diverse Perspectives** – “Different vision, different opinion” builds confidence in original thought.
- **Growth Stories** – Share real examples of persistence from past student work or personal experience.
- **“Lifelong Learner” Modeling** – Admit what you don’t know and learn alongside students.

● *Purpose:* Normalize persistence as a shared, ongoing habit of mind.

Teacher Moves

- Model curiosity, patience, and self-compassion during challenges.
- Provide time and structure for iteration — don’t rush the process.

- Frame feedback around effort, revision, and next steps.
- Celebrate perseverance in class rituals and reflections.
- Teach that persistence is not just endurance — it's adaptive, reflective, and hopeful.

Disciplined

Creative learners refine, revise, and reflect.

Discipline in creative thinking is about developing skill and craftsmanship over time — knowing when to take risks and when to slow down for reflection and refinement. It's the steady practice of turning ideas into excellence through iteration, feedback, and care. Teachers cultivate disciplined thinking by building habits of revision, scaffolding complex skills, and guiding students to think critically about their process and results.

"When you think you're done, you've just begun."

What It Looks Like

- Students approach creativity as a process of continual refinement.
 - Feedback, revision, and reflection are embedded in daily learning.
 - Teachers make craftsmanship, precision, and clarity visible and attainable.
 - Students see their work as prototypes — always in progress toward improvement.
-

Practices to Try

1. Drafting, Revising, and Refining

Students learn that high-quality work comes through iteration, not instant perfection.

- **Re-draft and Edit** – Multiple drafts of writing, art, or design projects; revision as a standard expectation.

- **“Roll the Dice” Revision Process** – Randomly choose areas of focus (e.g., word choice, structure, visuals) for targeted improvement.
- **Prototype Modeling** – Label and reflect on version 1, 2, 3, etc., to make progress visible.
- **Printmaking for Social Justice** – Sketch, transfer, engrave, proof, and revise; peer feedback at every step.
- **Engineering & BIG Lab Design Process** – Build, test, fail, and improve through iterative cycles.
- **Treat Work as a Prototype** – Discuss “What would we change in post?” before calling it finished.
- **Revising in Contexts Beyond Writing** – Return to a math problem or science design at the end of a unit to see new strategies or insights.

● *Purpose:* Help students internalize that craftsmanship grows through continual revision and attention to detail.

2. Reflection and Self-Assessment

Disciplined thinkers step back to evaluate their process, effort, and results.

- **Self-Assessment Rubrics** – Students identify what’s strong and what to improve before turning in work.
- **Analyze Exemplars** – Evaluate anonymous or teacher-created samples to recognize quality.
- **“Are You Really Done?” Poster** – Guiding questions prompt students to recheck, revise, and refine.
- **End-of-Semester Reading Log Review** – Students analyze patterns and growth in reading choices.
- **Reflection Journals** – Students record what went well, what was challenging, and what to try next.

- **Success Criteria Checklists** – Students co-create checklists, trade with peers, and use them to self-monitor progress.
- **Revisit Feedback** – After revision, return to rubric comments to assess improvement.

● *Purpose:* Foster independent learners who can critically evaluate and improve their own work.

3. Precision, Craft, and Technical Mastery

Discipline involves the art of skill — focusing on clarity, accuracy, and technique.

- **Modeling Craftsmanship** – Show actions slowly and narrate the “why” behind each choice.
- **Choosing the Right Tool for the Job** – Scaffold decision-making about materials, methods, or strategies.
- **Mathematical Explanation and Defense** – Students justify and “prove” their reasoning in writing and discussion.
- **Evaluate the Mistake** – Study errors to identify what could be improved and why.
- **Fair Test Science Challenges** – Test, adjust, and refine experiments systematically.
- **Group Science Projects (Simple Machines)** – Trial, error, and multiple redesigns to achieve a functional model.
- **Public Presentations** – Presenting to a professional audience encourages high standards and attention to detail.

● *Purpose:* Teach that mastery is not rote — it’s thoughtful, practiced, and purposeful.

4. Scaffolding and Step-by-Step Learning

Structure gives creative growth a path forward.

- **Bite-Sized Steps** – Break complex tasks into manageable parts (first, then, next, last).
- **Iteration as Routine** – Make revision cycles part of normal classroom rhythm.
- **Encouragement and Materials Adjustments** – Offer alternative tools or approaches when students hit barriers.
- **“Broad to Narrow” Brainstorming** – Begin expansive, then winnow ideas down to the essential.
- **Multiple Task Attempts** – Repeat a skill in varied ways to analyze effectiveness and personal preference.
- **Cocreate Rubrics and Success Criteria** – Involve students in defining what “quality” looks like.
- **Time-Bound Work Blocks** – Fixed work times (e.g., “no early turn-ins”) reinforce sustained focus.

● *Purpose:* Balance creativity with structure — providing scaffolds that lead to independence.

5. Reflection Through Feedback and Collaboration

Students refine thinking when they exchange ideas and critique with peers.

- **Peer Review & Editing** – Students help each other “make it better, make it connect.”
- **Group Strategy Sharing** – Students explain personal strategies and listen to others’.
- **Collaborative Rubric Design** – Students help define assessment language, deepening ownership.
- **Math Reflection Prompts** – “Can we read someone’s work and understand their thinking?”
- **Partner Check-Ins** – Structured feedback rounds (“Is it ready to go?” “What could change?”).
- **Feedback Stories** – Teachers share examples, like “Your writing is not gold,” to illustrate the power of revision and detachment from one’s first draft.

● *Purpose:* Create a culture where critique is kind, specific, and essential to mastery.

6. Iteration and Reflection Beyond the Classroom

Disciplined creativity extends to real-world and cross-curricular contexts.

- **Public Showcases or Panels** – Students present refined work to authentic audiences.
- **Documentation of Learning** – Students trace their process through photos, journals, or portfolios.
- **Reflections on Predictions vs. Outcomes** – Compare what was expected to what happened and why.
- **STEM Challenges and Prototyping** – Test, observe, and adjust designs using real feedback loops.
- **Revisiting Ideas Over Time** – Encourage “returning to unfinished work” as a regular practice.

● *Purpose:* Build habits of professionalism, precision, and reflective pride in craftsmanship.

Teacher Moves

- Model revision openly — show that your first draft rarely stays your last.
- Embed reflection and self-assessment into every project cycle.
- Use visible tools (rubrics, checklists, posters) to scaffold discipline and independence.
- Celebrate craftsmanship and improvement more than speed or completion.
- Encourage curiosity about *how* improvement happens — not just *that* it does.

Inquisitive

Creative learners wonder, investigate, and ask why.

Inquisitive thinking begins with curiosity — a desire to understand how things work and why they are the way they are. Students explore, question, and challenge assumptions to uncover new perspectives. Teachers nurture inquisitiveness by creating space for wonder, modeling their own questioning, and encouraging students to see learning as a series of discoveries rather than answers.

"Curiosity is the compass of creativity."

What It Looks Like

- Students ask deep, open-ended questions and pursue their own lines of inquiry.
 - Teachers model curiosity and encourage "why" and "what if" questions.
 - Classrooms embrace uncertainty and exploration over correctness.
 - Investigations start from noticing, wondering, and pattern-seeking.
-

Practices to Try

1. Wonder as a Starting Point

Help students see questions as invitations, not obstacles.

- **What Do You Notice? What Do You Wonder?** – Use with images, text, math problems, or phenomena.
- **Wonder Charts / Wonder Walls** – Collect ongoing questions across subjects ("What do you want to know more about in reading, in math, in science?").
- **KWL Charts** – Activate curiosity by identifying what students *Know*, *Want to know*, and later *Learned*.
- **Introducing a Unit with a Mystery** – Show a surprising photo or demo; ask, "What's happening here?"

- **Phenomenon First** – Present an unexplained event or pattern before teaching the content.
- **Prediction Prompts** – “What do you think will happen?” or “What makes you say that?”

● *Purpose:* Make wonder visible and central to learning.

2. Question-Based Routines

Structured routines turn curiosity into investigation.

- **Here’s the Answer, What’s the Question?** – Students generate multiple questions for a single answer (49, tragedy, etc.).
- **Visual Thinking Strategies** – Ask: “What’s going on in this picture?” “What do you see that makes you say that?” “What more can we find?”
- **See-Think-Wonder** – A simple, powerful framework for analyzing visuals, texts, or data.
- **Picture Analysis (See / Assume / Wonder)** – Students distinguish between observation and inference.
- **A Question Is the Answer** – Students respond to every prompt with a question.
- **Post-it Matching** – Pairing questions and answers to encourage critical connection-making.
- **“Is This Always True?”** – Invite exploration of exceptions and conditions.

● *Purpose:* Build disciplined curiosity — learning to question with depth and precision.

3. Investigating and Proving Thinking

Encourage inquiry that goes beyond recall to reasoning and justification.

- **Math Inquiries** – “If this is the answer, what’s the question?” or “How many ways can we arrive here?”

- **Checking the Teacher Edition** – Students evaluate multiple solution paths and check for differences.
- **Disagree and Prove It** – Students respectfully challenge a solution and explain their reasoning.
- **Backwards Problem Solving** – Give results first and let students determine the steps that lead there.
- **Engineering Design Inquiry** – Students prototype, test, and revise (e.g., designing a back scratcher with the BIG Lab).
- **“How Did I Get There?”** – Students analyze and verbalize their process to reveal hidden assumptions.

● *Purpose:* Position students as investigators who construct, test, and refine understanding.

4. Challenging Assumptions and Exploring Perspectives

Curiosity deepens when students see the world from new angles.

- **Perspective-Taking** – Explain three different viewpoints on the same issue or problem.
- **Comparing Cultures Through Images** – “What’s the same? What’s different?”
- **Gender and Identity Inquiry** – Read books with characters who challenge labels; discuss why we want to define them.
- **Philosophical Questions** – “Are your dreams real?” “Is this always true?”
- **Inquiry-Based Discussion in Literature** – Ask, “How does your interpretation change when you hear someone else’s?”
- **Cognitive Dissonance** – Introduce edge cases that challenge established rules or definitions.

● *Purpose:* Encourage intellectual humility and curiosity about complexity.

5. Playful and Digital Inquiry

Curiosity thrives when learning is interactive, exploratory, and open-ended.

- **Wikipedia Surfing** – Start on one topic and navigate to a target topic through connected hyperlinks.
- **Materials Exploration** – Offer varied tools or media and invite students to “see what happens.”
- **Math and Logic Challenges** – Use open-ended or playful prompts (“Use the digits 1–9 once to make two fractions whose product is $\frac{3}{4}$ ”).
- **Post-it Question Swaps** – Students circulate and add questions or answers to others’ prompts.
- **Inquiry Scavenger Hunts** – Give partial information and ask students to fill in missing context.

 *Purpose:* Connect curiosity with play and discovery, encouraging flexible exploration.

6. Modeling and Sustaining Curiosity

Teachers’ own wondering sets the tone for inquiry-based learning.

- **Model Your Own Inquisitive Process** – Think aloud: “I looked at this data and I’m wondering...”
- **Ask Students to Predict Questions** – “What’s one question someone might ask about this?”
- **Always Ask “Why?” or “What Makes You Say That?”** – Push students to explain reasoning.
- **Encourage “I See” Statements** – Ground interpretations in observation before opinion.
- **End with Unanswered Questions** – Use cliffhangers or unsolved problems to spark future learning.

- **Science Symposiums and Open Forums** – Invite students to present investigations and field peer questions.

● *Purpose:* Model lifelong learning — curiosity as a practice, not a phase.